

BRSM Movie Memory Experiment

Comprehensive Statistical Report
After Vigilance-Based Participant Exclusions

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1 Overview and Data Quality

1.1 Vigilance-based exclusions

Two participants failed the embedded vigilance check and are excluded from every analysis:

- sub105_AB
- sub70_NB

The retained analytical sample is **N = 168** (NB = 88, AB = 80), contributing **6,720 recognition trials**.

1.2 Sample and trial structure

Table 1: Analytical sample after vigilance exclusions.

Condition	<i>n</i> participants	<i>n</i> trials	BB trials	EM trials
NB (Natural Boundary)	88	3,520	1,760	1,760
AB (Abrupt Boundary)	80	3,200	1,600	1,600
Total	168	6,720	3,360	3,360

2 Statistical Methods

Each hypothesis follows an identical pre-specified decision pipeline:

1. Build the DV at the participant level from trial-level data.
2. Compute full descriptive statistics (*n*, *M*, *SD*, *Mdn*, *Q*₁, *Q*₃, min, max) per group.
3. Test normality within each group using **Shapiro–Wilk**.
4. Test variance homogeneity using **Levene’s test, median-centred** (= Brown–Forsythe).
5. Select the test:
 - Both groups normal *and* equal variance ⇒ **Student’s *t*-test**
 - Both groups normal *and* unequal variance ⇒ **Welch’s *t*-test**
 - Either group non-normal ⇒ **Mann–Whitney *U* test** (two-sided)
6. Report effect sizes: Cohen’s *d* / Hedges’ *g* for parametric tests; rank-biserial *r* for non-parametric tests.
7. No correction for multiple comparisons is applied. All reported *p*-values are raw.

2.1 Accuracy dependent variable

All accuracy-based DVs use **raw proportion correct** (range 0–1). Chance performance is 0.5 in a two-alternative forced-choice task.

2.2 Log-transformation and normality of response time

Trial-level RTs are strongly right-skewed (skew ≈ 3.83 , excess kurtosis ≈ 29). All RT analyses therefore use **natural-log-transformed RT** (ln seconds). Table 2 and Figures 1 and 2 verify that the transform rescues approximate normality at the participant level.

Table 2: Shapiro–Wilk normality: raw RT vs. log RT at participant level.

Condition	Transform	n	M	Shapiro–Wilk p
NB	Raw RT (s)	88	5.816 s	< .001 (non-normal)
	Log RT (ln s)	88	1.611	.5957 (normal)
AB	Raw RT (s)	80	5.569 s	< .001 (non-normal)
	Log RT (ln s)	80	1.572	.0685 (normal)
All (trial)	Raw RT	5,000 sample	skew = 3.829	< .001
	Log RT	5,000 sample	skew = 0.618	< .001

Note. At the trial level ($n \approx 6720$) Shapiro–Wilk has extreme power and flags even trivial departures from normality. The skew drops from ≈ 3.83 (raw) to ≈ 0.62 (log), confirming substantial normalisation. At the participant level both conditions pass after log-transformation.

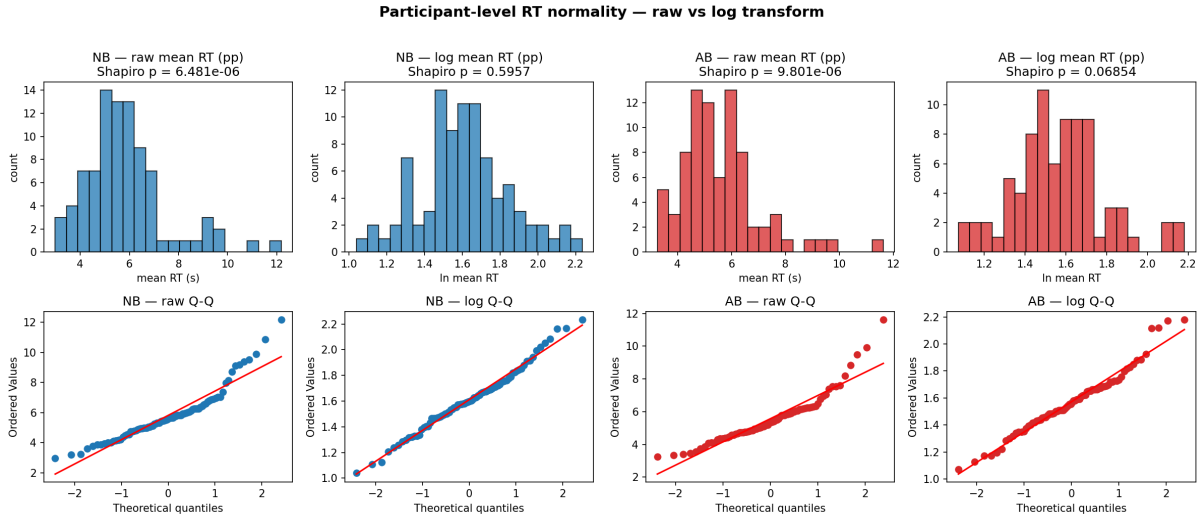


Figure 1: Participant-level RT distributions: raw (columns 1 and 3) vs. log-transformed (columns 2 and 4), for NB (left pair) and AB (right pair). Top row: histograms; bottom row: Q–Q plots.

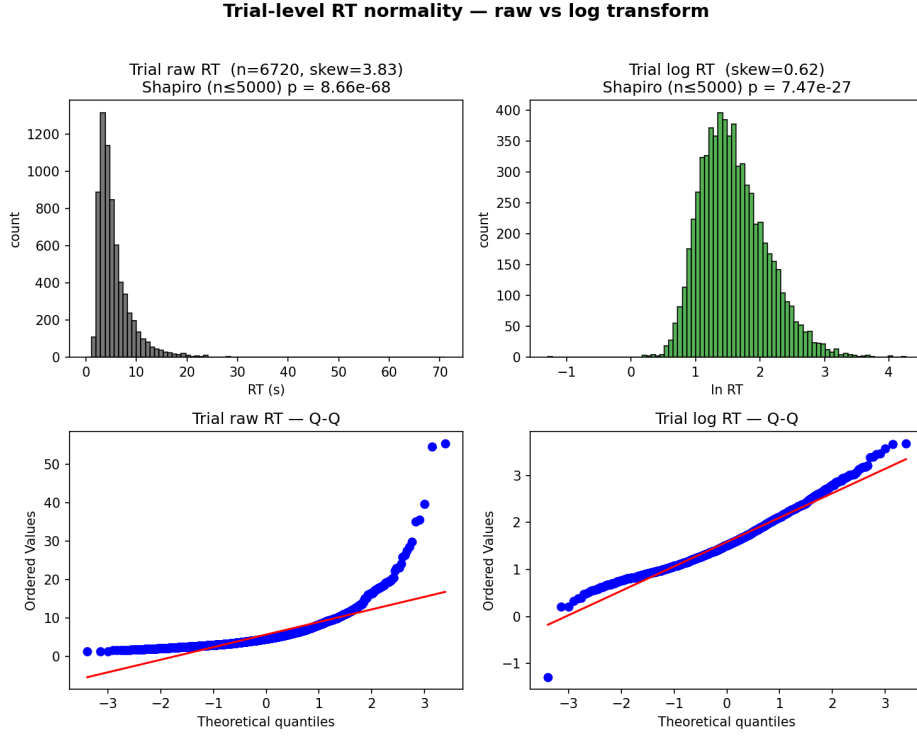


Figure 2: Trial-level RT distributions. Raw RT (top left) is strongly right-skewed; log RT (top right) is approximately symmetric. Q-Q plots (bottom) confirm the improvement.

3 H1: Overall Recognition Accuracy

3.1 Hypotheses

H_0 : Overall recognition accuracy does not differ between Natural Boundary and Abrupt Boundary participants ($\mu_{NB} = \mu_{AB}$).

H_1 : NB participants have higher overall recognition accuracy than AB participants ($\mu_{NB} > \mu_{AB}$; directional).

3.2 DV and descriptive statistics

DV: Per-participant proportion correct across all 40 recognition trials.

Table 3: Descriptive statistics — H1 (Overall accuracy).

Group	<i>n</i>	<i>M</i>	<i>SD</i>	<i>Mdn</i>	<i>Q</i> ₁	<i>Q</i> ₃	Min	Max
NB	88	0.871	0.073	0.875	0.825	0.925	0.650	1.000
AB	80	0.841	0.082	0.850	0.794	0.900	0.575	0.975

3.3 Test selection and result

Normality (Shapiro–Wilk): NB $p = < .001$; AB $p = .0042$. Both groups reject normality (ceiling-bounded distribution).

Variance homogeneity (Levene): $p = .3412$.

Test selected: Mann–Whitney U (both groups non-normal).

Result: $U = 4297.0$, $p_{\text{two-sided}} = .0130$, $p_{\text{one-sided}} = .0065$. Rank-biserial $r_{\text{rb}} = 0.221$.

Decision: Reject H_0 . NB participants have significantly higher overall accuracy.

4 H2: Before-Boundary Frame Accuracy

4.1 Hypotheses

H_0 : Recognition accuracy for Before-Boundary targets does not differ between conditions ($\mu_{\text{NB}} = \mu_{\text{AB}}$).

H_1 : NB participants have higher Before-Boundary accuracy than AB participants ($\mu_{\text{NB}} > \mu_{\text{AB}}$; directional).

4.2 DV and descriptive statistics

DV: Per-participant proportion correct on BB-frame targets only (20 trials).

Table 4: Descriptive statistics — H2 (Before-Boundary accuracy).

Group	n	M	SD	Mdn	Q_1	Q_3	Min	Max
NB	88	0.860	0.097	0.900	0.800	0.950	0.600	1.000
AB	80	0.826	0.104	0.850	0.750	0.900	0.450	1.000

4.3 Test selection and result

Normality (Shapiro–Wilk): NB $p = < .001$; AB $p = .0015$. Both reject normality.

Variance homogeneity (Levene): $p = .7323$.

Test selected: Mann–Whitney U .

Result: $U = 4249.0$, $p_{\text{two-sided}} = .0190$, $p_{\text{one-sided}} = .0095$. $r_{\text{rb}} = 0.207$.

Decision: Reject H_0 . NB participants recognise BB-frame targets significantly more accurately.

5 H3: Mean Confidence Rating

5.1 Hypotheses

H_0 : Mean confidence rating does not differ between NB and AB participants ($\mu_{\text{NB}} = \mu_{\text{AB}}$).

H_1 : NB participants report higher mean confidence than AB participants ($\mu_{\text{NB}} > \mu_{\text{AB}}$; directional).

5.2 DV and descriptive statistics

DV: Per-participant mean confidence rating (1–5 Likert scale).

Table 5: Descriptive statistics — H3 (Mean confidence rating).

Group	n	M	SD	Mdn	Q_1	Q_3	Min	Max
NB	88	4.207	0.412	4.275	3.969	4.513	3.000	5.000
AB	80	4.087	0.457	4.100	3.700	4.431	2.900	4.900

5.3 Test selection and result

Normality (Shapiro–Wilk): NB $p = .0364$; AB $p = .1371$. NB rejects normality (ceiling effect — most ratings ≥ 4).

Variance homogeneity (Levene): $p = .1496$.

Test selected: Mann–Whitney U .

Result: $U = 4050.5$, $p_{\text{two-sided}} = .0923$, $p_{\text{one-sided}} = .0461$. $r_{\text{rb}} = 0.151$.

Decision: Fail to reject H_0 . NB participants are numerically more confident, but the difference is not significant. A ceiling effect likely reduces power.

6 H4: Proportion of Low-Confidence Trials

6.1 Hypotheses

H_0 : The proportion of trials with confidence ≤ 3 does not differ between AB and NB participants ($\pi_{\text{AB}} = \pi_{\text{NB}}$).

H_1 : AB participants have a higher proportion of low-confidence trials than NB participants ($\pi_{\text{AB}} > \pi_{\text{NB}}$; directional).

6.2 DV and descriptive statistics

DV: Per-participant proportion of trials rated confidence ≤ 3 .

Table 6: Descriptive statistics — H4 (Proportion of low-confidence trials).

Group	n	M	SD	Mdn	Q_1	Q_3	Min	Max
AB	80	0.263	0.147	0.263	0.150	0.375	0.000	0.650
NB	88	0.233	0.127	0.225	0.144	0.300	0.000	0.625

6.3 Test selection and result

Normality (Shapiro–Wilk): AB $p = .1510$; NB $p = .0453$.

Test selected: Mann–Whitney U test (two-sided, continuity-corrected).

Result: $U = 3918.500$, $p_{\text{two-sided}} = .2053$, $p_{\text{one-sided}} = .1027$. Effect: Rank-biserial $r = 0.113$.

Decision: Fail to reject H_0 . AB participants show numerically more low-confidence responses, but the difference is not significant.

7 H5: Per-Movie Accuracy Variability

7.1 Hypotheses

H_0 : The standard deviation of per-movie accuracy does not differ between AB and NB participants ($\sigma_{\text{AB}} = \sigma_{\text{NB}}$).

H_1 : AB participants show greater variability in per-movie accuracy than NB participants ($\sigma_{\text{AB}} > \sigma_{\text{NB}}$; directional).

7.2 DV and descriptive statistics

DV: Per-participant standard deviation of accuracy across the 40 movies.

Table 7: Descriptive statistics — H5 (Per-movie accuracy variability).

Group	n	M	SD	Mdn	Q_1	Q_3	Min	Max
AB	80	0.353	0.079	0.362	0.304	0.410	0.158	0.501
NB	88	0.320	0.086	0.335	0.267	0.385	0.000	0.483

7.3 Test selection and result

Normality (Shapiro–Wilk): AB $p = .0069$; NB $p = .0032$. Both reject normality.

Variance homogeneity (Levene): $p = .6271$.

Test selected: Mann–Whitney U .

Result: $U = 4337.0$, $p_{\text{two-sided}} = .0093$, $p_{\text{one-sided}} = .0047$. $r_{\text{rb}} = 0.232$.

Decision: Reject H_0 . AB participants show significantly greater per-movie accuracy variability.

8 H6: BB-Frame vs EM-Frame Response Time

8.1 Hypotheses

H_0 : Mean log RT does not differ between BB-frame and EM-frame recognition trials ($\mu_{\text{BB}} = \mu_{\text{EM}}$).

H_1 : Mean log RT differs between BB-frame and EM-frame recognition trials ($\mu_{\text{BB}} \neq \mu_{\text{EM}}$; non-directional).

Rationale. If BB-frame targets engage a qualitatively different retrieval process (e.g., recollection rather than familiarity), recognition decisions for BB targets may take longer than for EM targets. This within-participant comparison is exploratory and non-directional.

8.2 DV and descriptive statistics

DV: Per-participant mean log RT on BB trials and mean log RT on EM trials; the within-participant difference BB – EM is tested against zero.

Table 8: Descriptive statistics — H6 (BB vs EM log RT, within-participant).

Measure	n	M	SD	Mdn	Q_1	Q_3	Min	Max
BB mean log RT (ln s)	168	1.610	0.240	1.582	1.473	1.735	1.048	2.289
EM mean log RT (ln s)	168	1.575	0.242	1.583	1.413	1.709	1.008	2.224
BB – EM difference	168	0.035	0.134	0.037	-0.056	0.133	-0.328	0.355

8.3 Test selection and result

Normality of differences (Shapiro–Wilk): $p = .5319$ (normal). **One-sample t -test** on differences is used.

Result: $t = 3.378$, $df = 167$, $p_{\text{two-sided}} = < .001$. Effect: Cohen’s $d_z = 0.261$.

Decision: Reject H_0 . BB-frame recognition decisions take longer than EM-frame decisions (mean difference = 0.035 ln s), consistent with a more effortful retrieval process for boundary-region events.